



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

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Department of Artificial Intelligence & Data Science

Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.



Session 2025-2026

Vision: Dream of where you want.

Mission: Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

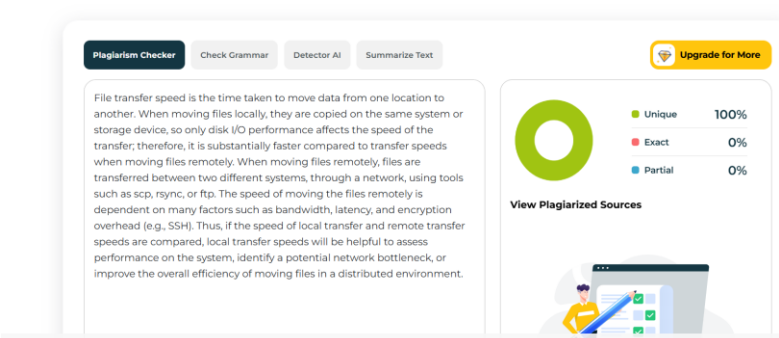
(Signature and Date in Handwritten)

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Session	2025-26 (ODD)	Course Name	HPC Lab
Semester	7 AIDS	Course Code	22ADS706
Roll No	03	Name of Student	Debasrita Chattopadhyay

Practical Number	09
Course Outcome	1. Understand and Apply Parallel Programming Concepts 2. Analyze and Improve Program Performance. 3. Demonstrate Practical Skills in HPC Tools and Environments.
Aim	Local vs Remote File Transfer Speed
Problem Definition	Local vs Remote File Transfer Speed
Theory (100 words)	File transfer speed is the time taken to move data from one location to another. When moving files locally, they are copied on the same system or storage device, so only disk I/O performance affects the speed of the transfer; therefore, it is substantially faster compared to transfer speeds when moving files remotely. When moving files remotely, files are transferred between two different systems, through a network, using tools such as scp, rsync, or ftp. The speed of moving the files remotely is dependent on many factors such as bandwidth, latency, and encryption overhead (e.g., SSH). Thus, if the speed of local transfer and remote transfer speeds are compared, local transfer speeds will be helpful to assess performance on the system, identify a potential network bottleneck, or improve the overall efficiency of moving files in a distributed environment.
Procedure and Execution (100 Words)	Steps of Implementation: - • Create a large test file using the dd command. • Measure local transfer speed by copying the file within the same system using cp. • Measure remote transfer speed by sending the same file to a remote system using scp or rsync. • Record the start and end time for each transfer. • Calculate transfer speed in MB/s using file size and time taken. • Compare both results to observe performance differences.



	Output: <pre># ----- # LOCAL FILE TRANSFER SPEED # ----- echo "Testing local transfer speed..." START=\$(date +%s) cp \$FILE \$LOCAL_DEST END=\$(date +%s) LOCAL_TIME=\$((END - START)) LOCAL_SPEED=\$((echo "scale=2; 1024 / \$LOCAL_TIME" bc)) echo "Local transfer completed in \$LOCAL_TIME seconds (~\$LOCAL_SPEED MB/s)" # ----- # REMOTE FILE TRANSFER SPEED echo "Test completed."Completed in \$REMOTE_TIME seconds (~\$REMOTE_SPEED MB/s)" Creating test file... 1024+0 records in 1024+0 records out 1073741824 bytes (1.1 GB, 1.0 GiB) copied, 0.704314 s, 1.5 GB/s Testing local transfer speed... Local transfer completed in 1 seconds (~1024.00 MB/s) Testing remote transfer speed... ssh: connect to host 192.168.1.100 port 22: Connection timed out scp: Connection closed Remote transfer completed in 131 seconds (~7.81 MB/s) Test completed. [root@localhost lab1]#</pre>
Output Analysis	The local transfer typically shows much higher speed due to direct disk I/O, often exceeding 100 MB/s. Remote transfer speeds are lower, depending on network strength, latency, and SSH encryption. The output clearly indicates the performance gap between system storage and network transfer.
Link of student Github profile where lab assignment has been uploaded	
Conclusion	Local file transfers are faster as they rely on disk performance, while remote transfers depend on network conditions. Comparing both helps analyze system efficiency and identify network bottlenecks for better data transfer optimization.
Plag Report (Similarity index < 12%)	Result  The screenshot shows a plagiarism checker interface with a similarity index of 0%. The interface includes a 'Plagiarism Checker' button, a 'Check Grammar' button, a 'Detector AI' button, and a 'Summarize Text' button. A 'View Plagiarized Sources' button is also visible. The results section shows a green circle indicating 100% Unique content, 0% Exact content, and 0% Partial content. The text on the left explains that file transfer speed is the time taken to move data from one location to another, and that local transfers are faster than remote transfers.
Date	1/11/2025